

9. (a) (i) Explain the origins of emission spectra. [2]

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- (ii) Give **one** difference in the appearance of absorption and emission spectra. [1]

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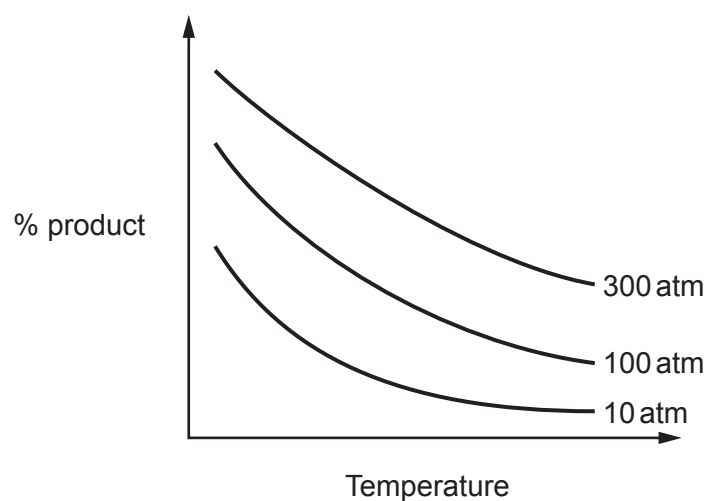
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- (b) A line in the emission spectrum of an element has a wavelength of 95.0 nm. Calculate the frequency of this line in megahertz, MHz. [2]

frequency = MHz



- (c) The diagram below shows how the percentage product varies with temperature and pressure for an equilibrium process.



Use the diagram and Le Chatelier's principle to explain whether:

- (i) the forward reaction is endothermic or exothermic.

[2]

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- (ii) the forward reaction involves an increase or decrease in the number of moles of gas.

[2]

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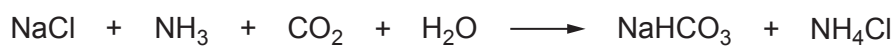
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- (d) The Solvay process is the major industrial process for the manufacture of sodium carbonate. Two of the stages in the process are shown below:



Calculate the maximum mass of sodium carbonate, in kg, which could be obtained from 15.0 tonnes of sodium chloride. Give your answer to an appropriate number of significant figures.

[3]

maximum mass = kg

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